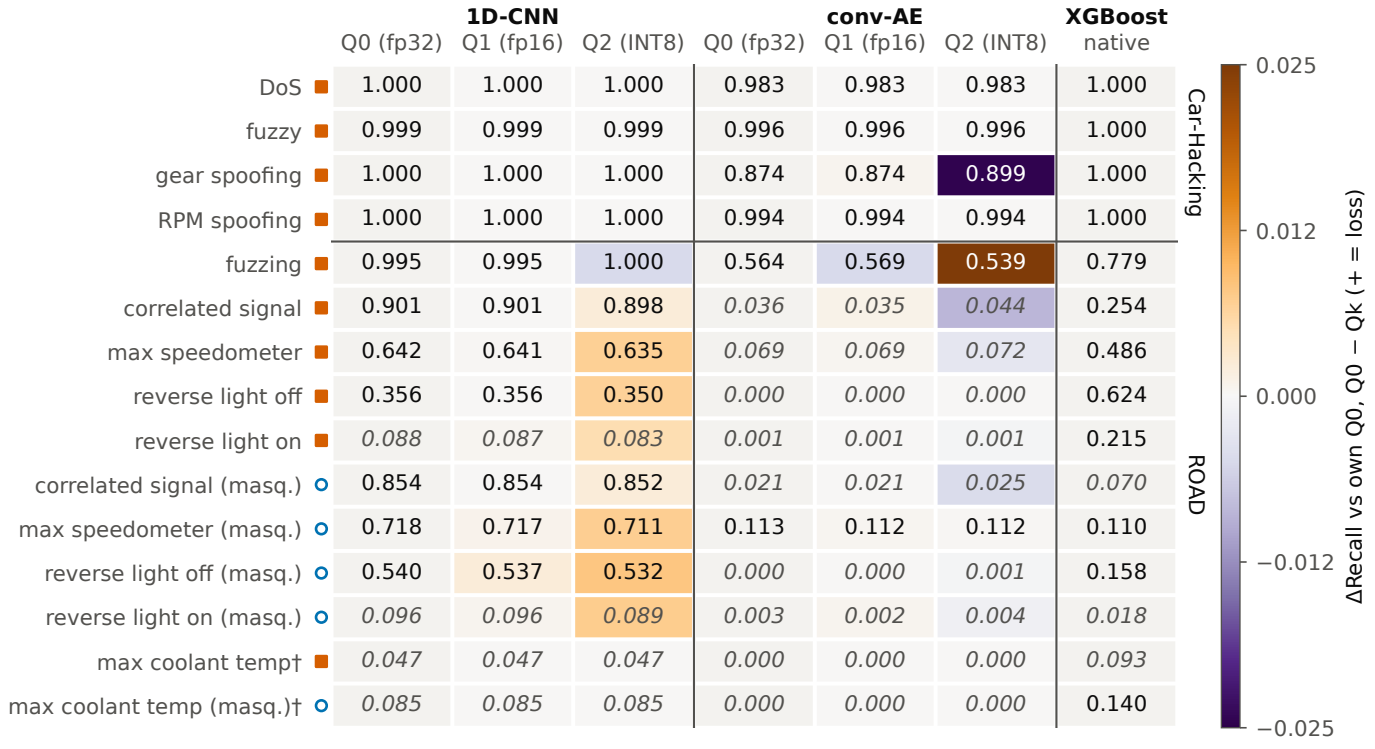
1D-CNN
conv-AE
XGBoost
Q0 (fp32)
Q1 (fp16)
Q2 (INT8)
Q0 (fp32)
Q1 (fp16)
Q2 (INT8)
native
DoS 1.000 1.000 1.000 0.983 0.983 0.983 1.000
fuzzy 0.999 0.999 0.999 0.996 0.996 0.996 1.000
gear spoofing 1.000 1.000 1.000 0.874 0.874 0.899 1.000
RPM spoofing 1.000 1.000 1.000 0.994 0.994 0.994 1.000
fuzzing 0.995 0.995 1.000 0.564 0.569 0.539 0.779
correlated signal 0.901 0.901 0.898 0.036 0.035 0.044 0.254
max speedometer 0.642 0.641 0.635 0.069 0.069 0.072 0.486
reverse light off 0.356 0.356 0.350 0.000 0.000 0.000 0.624
reverse light on 0.088 0.087 0.083 0.001 0.001 0.001 0.215
correlated signal (masq.) 0.854 0.854 0.852 0.021 0.021 0.025 0.070
max speedometer (masq.) 0.718 0.717 0.711 0.113 0.112 0.112 0.110
reverse light off (masq.) 0.540 0.537 0.532 0.000 0.000 0.001 0.158
reverse light on (masq.) 0.096 0.096 0.089 0.003 0.002 0.004 0.018
max coolant temp† 0.047 0.047 0.047 0.000 0.000 0.000 0.093
max coolant temp (masq.)† 0.085 0.085 0.085 0.000 0.000 0.000 0.140
Car-Hacking
ROAD
0.025
0.012
0.000
−0.012
−0.025
ΔRecall vs own Q0, Q0 − Qk (+ = loss)